

```
class MDPsim:
```

```
def __init__(self, s, r):
    self.s = s
    self.r = r
```

```
def acciones_legales(self, s):
    return iterable con acciones legales en s
```

```
def sucesor(self, s, a):
    return s'
```

```
def recompensas(self, s, a, s'):
    return r
```

```
def terminal(self, s):
    return True/False
```

```
def estado_inicial(self):
    return estado
```

```
class Politica:
```

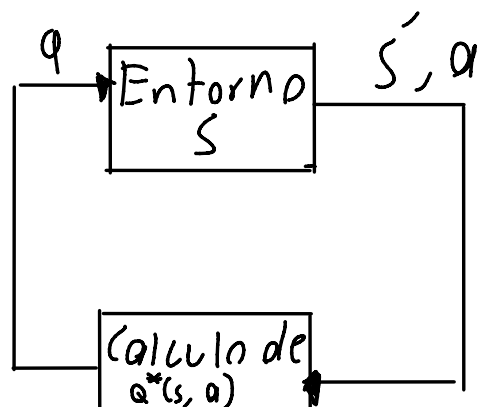
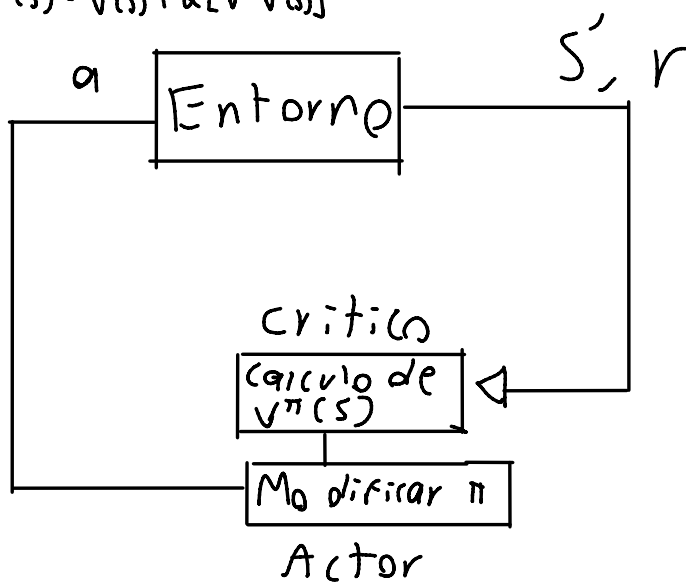
```
def __init__(self, p_dict):
    self.pi = p_dict
```

```
def __call__(self, s):
    return self.pi[s]
```

$$V^{\pi}(s) = \sum_{a \in A(s)} \pi(s, a) \sum_{s'} T(s, a, s') [r(s, a, s') + \gamma V^{\pi}(s')]$$

Yo tengo $\hat{V}^{\pi}(s)$ para un estado s
 aplico

```
q = politica(s)
s' = MDPsim.sucesor(s, a)
r = MDPsim.ganancia(s, a, s')
v = r + MDPsim.Y * V^pi(s')
V^pi(s) = V^pi(s) + alpha [v - V^pi(s)]
```



```
def TD_0(mdl,  $\pi$ ,  $\alpha$ , max_episodios, max_it, tol)
```

```
    V = {0 for s in mdl.S}
```

```
    for _ in range(max_episodios):
```

```
        err = 0
```

```
        mdl = modelo
```

```
         $\pi$  = politica
```

```
        s = mdl.estado_inicial()
```

```
        tol = tolerancia
```

```
        for _ in range(max_it):
```

```
            a =  $\pi$ (s)
```

```
            s' = mdl.sucesor(s, a)
```

```
            r = mdl.recompensa(s, a, s')
```

```
            V = V[s] -  $\alpha \cdot (r - \text{mdl. } \gamma \cdot V[s'] - V[s])$ 
```

```
            err = max(err, abs(V - V[s]))
```

```
            V[s], s = V, s'
```

```
            if mdl.termina(s):
```

```
                break
```

```
        if err < tol:
```

```
            break
```

```
    return V
```

$$\pi(s) = \arg \max_{a \in A_L(s)} Q^*(s, a)$$

$$Q^*(s, a) = \sum_{s' \in S} T(s, a, s') [r(s, a, s') + \gamma \sum_{a' \in A_L(s')} \pi(s', a') Q^*(s', a')]$$

$$Q^*(s, a) = \sum_{s' \in S} T(s, a, s') [r(s, a, s') + \gamma \max_{a' \in A_L(s')} Q^*(s', a')]$$

Tengo $Q(s, a)$

aplico:

```
s' = mdl.sucesor(s, a)
```

```
r = mdl.recompensa(s, a, s')
```

```
(a' = arg max_a Q(s', a'))
```

```
(a' = politica(s'))
```

```
q = r + mdl.  $\gamma \cdot Q(s', a')$ 
```

```
Q(s, a) = Q(s, a) +  $\alpha (q - Q(s, a))$ 
```